Anshul Raje

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US Citizen | Seeking Summer 2027 Internship | Robotic Perception & Computer Vision

EDUCATION

M.Sc. in Robotics Engineering, Worcester Polytechnic Institute, Worcester, MA — **Aug 2026 – May 2028**
Courses: Vision-based Robotic Manipulation, Hands-on Autonomous Aerial Robotics

B.E. in Electronics and Communication Engineering, BITS Pilani, Goa, India — **Sep 2020 – Jul 2024**
GPA: 3.40/4.0 | Courses: Digital Image Processing, Control Systems, Digital Signal Processing

SKILLS

Languages:	Python, C/C++
Frameworks & Libraries:	TensorFlow, PyTorch, ROS 2, ROS 2 Control, MoveIt, OpenCV, PCL, Ultralytics
Tools & Platforms:	Linux, Git, Docker, Gazebo, RViz, Blender, ONNX Runtime

EXPERIENCE

Strider Robotics | Associate Robotics Engineer | Bangalore, India — **Aug 2024 – Apr 2026**

- Automated analog gauge reading by detecting needle base, needle tip, and the min and max scale markers, then converting the angle between them into a calibrated value, reading 22 of 25 gauges correctly per run on site.
- Built a vision-guided capture pipeline servoing a PTZ camera onto a known ArUco marker and then onto the gauge itself, zooming and autofocusing to image each asset from a repeatable viewpoint every round.
- Cut inference cost in the alignment loop by running YOLO with DeepSORT tracking only on the first pass, then switching to normalized cross-correlation template matching for every later alignment step.
- Trained YOLOv12 detectors on 2500+ hand-labelled images spanning gauge, needle, scale marker, and fire extinguisher classes, converting them to TensorRT engines to run 720p inference at 25 FPS onboard a Jetson.
- Designed a five-layer ROS 2 perception architecture separating camera drivers, computer vision inference nodes, and task orchestration, and deployed it on legged inspection robots at two industrial clients.
- Architected a behavior tree orchestration framework separating reusable perception primitives from client-specific task sequences, letting the team add a new client's inspection routine as a configuration package in one to two hours.
- ***Tools & Techniques:*** *Visual Servoing, ArUco Marker Detection, Multi-Object Tracking, Template Matching, TensorRT, Edge Inference, Behavior Trees, DeepSORT.*

HiPeRT Lab, UNIMORE | Robotics Research Intern | Modena, Italy — **Jul 2023 – Dec 2023**

- Engineered a synthetic dataset pipeline segmenting track footage with Segment Anything Model (SAM) and compositing opponent car CAD models per frame, auto-annotating 15000+ images with zero manual labeling.
- Trained YOLOv4 on domain-randomized frames varying lighting, saturation, and blur, hitting a 97% detection rate on live on-track video and adding camera perception to a LiDAR-only race car stack.
- ***Tools & Techniques:*** *Synthetic Data Generation, Domain Randomization, Real-Time Inference, SAM, YOLOv4.*

PROJECTS

Perception-Driven Window Racing on a Quadrotor [Report] [Video] — **Oct 2026 – Nov 2026**

- Built a synthetic dataset pipeline in Blender rendering zero to three windows per image under randomized pose, lighting, texture, background, and camera noise, labeling 25000 images from scene geometry with zero manual labeling.
- Fine-tuned a YOLO26n-pose keypoint detector to locate the four inner corners of each window opening, reaching 0.967 precision and 0.933 recall on Gaussian splat renders of the course it never trained on.
- Estimated each window's 6-DoF pose by solving planar PnP on its detected corners, then fused repeated sightings in the world frame with RANSAC, mapping an unknown number of gates as the drone flew the course.
- Ran the detector on CPU through ONNX Runtime inside a 15 Hz state machine that searches for gates, plans to a line-up point with RRT*, and commits through each opening, replanning as gate estimates refine.
- Flew the stack hardware-in-the-loop on a Crazyflie under Vicon with camera images rendered from a Gaussian splat, clearing all three windows in 9.25 s on average at a 20 ms median control tick.
- ***Tools & Techniques:*** *Synthetic Data Generation, Domain Randomization, Blender, Keypoint Detection, YOLO26, 6-DoF Pose Estimation, Perspective-n-Point (PnP), RANSAC, ONNX Runtime, Edge Inference, State Machines, RRT*, Hardware-in-the-Loop.*

Multi-View Point Cloud Grasping on a Kinova Gen3 Lite [Report] [Video] **Oct 2026 – Nov 2026**

- Calibrated a wrist-mounted RealSense D405 to a Kinova Gen3 Lite by solving the eye-in-hand transform from ArUco marker poses, then captured point clouds from multiple viewpoints and fused them in the robot base frame.
- Segmented objects from the fused cloud by fitting the table plane with RANSAC and Euclidean clustering what remained, then selected antipodal grasps from surface normal pairs filtered by gripper width and table clearance.
- Executed pre-grasp, collision-aware approach, and lift through MoveIt on the real arm, picking up multiple objects with a 100% success rate inside the arm's reachable workspace.
- ***Tools & Techniques:*** *Point Cloud Processing, Hand-Eye Calibration, ArUco Marker Detection, Multi-View Registration, Iterative Closest Point (ICP), RANSAC Plane Segmentation, Euclidean Clustering, Antipodal Grasp Synthesis, MoveIt Motion Planning.*

Quadrotor Navigation with RRT* Planning and Quintic Trajectories [Report] [Video] **Sep 2026 – Oct 2026**

- Implemented an RRT* planner with goal-biased sampling, lowest-cost parent selection, and rewiring, then simplified its paths by greedy shortcutting against inflated obstacles to navigate a quadrotor through known 3D maps.
- Generated quintic spline trajectories continuous through snap by solving a block tridiagonal system for waypoint velocities and accelerations, scaling segment times to hold velocity and acceleration limits.
- Tuned the position and velocity loops of a cascaded PID controller to a mean tracking error of 14 cm in simulation, reaching the goal without collision in 500+ randomized runs across two maps.
- Flew a Crazyflie through a course of physical boxes under Vicon motion capture with a pure pursuit policy sending feedforward velocity commands, reaching the goal in 5.8 s with a 2.9 cm mean path error and no collision.
- ***Tools & Techniques:*** *Motion Planning, RRT*, Collision Checking, Trajectory Generation, Quintic Splines, Cascaded PID Control, Pure Pursuit, Quadrotor Simulation, Vicon Motion Capture, Sim-to-Real Transfer.*

Attitude Estimation using a 6-DoF IMU [Report] [Video] **Aug 2026 – Sep 2026**

- Benchmarked five attitude estimation methods against Vicon motion capture ground truth across six flight sequences, with an unscented Kalman filter reaching the lowest mean orientation RMSE of 14.6 degrees.
- Implemented a quaternion-based unscented Kalman filter fusing accelerometer data with a bias-calibrated gyroscope, propagating sigma points through the nonlinear process model to hold roll and pitch error to 5.4 and 4.8 degrees.
- ***Tools & Techniques:*** *State Estimation, Sensor Fusion, Unscented Kalman Filter (UKF), Madgwick Filter, Complementary Filter, Quaternion Attitude Representation, IMU Calibration, RMSE Benchmarking.*

Autonomous Navigation and Manipulation for a Mars Rover [Code] [Video] **Sep 2021 – May 2023**

- Added real-time obstacle detection to an autonomous rover by training YOLOv3 on a custom dataset, letting the rover identify and avoid hazards on course without operator input.
- Implemented a Stanley controller for autonomous waypoint navigation, holding tracking error within 5 cm across the traversal course.
- Developed forward and inverse kinematics for two joints of a 5-DOF onboard manipulator, letting operators command end-effector position for that pair instead of driving each joint separately.
- ***Tools & Techniques:*** *Object Detection, Real-Time Inference, Waypoint Navigation, Stanley Controller, Forward & Inverse Kinematics, YOLOv3.*

ACHIEVEMENTS

- Placed 2nd globally at the Anatolian Rover Challenge and won Best Autonomous and Controls (Jul 2022).
- Won the Excellence Award for Overall Performance at the International Rover Challenge, Bangalore (Jan 2023).